D3.1 Reproduction

NAME	DATE	TOTAL
		_____ / 51

About 50 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make. Look at the bold command term: "state" needs a brief answer, while "explain" needs reasons.

Section A: Recall

9 MARKS

1 **STATE** what is meant by a gamete. [D3.1.2] [1]

2 **STATE** one advantage of asexual reproduction. [D3.1.1] [1]

3 **STATE** where sperm are produced. [D3.1.4] [1]

4 **STATE** where fertilization usually takes place in humans. [D3.1.6] [1]

5 **STATE** the gland that secretes FSH and LH. [D3.1.5] [1]

6 **STATE** the hormone that triggers ovulation. [D3.1.5] [1]

7 **STATE** one function of the prostate gland. [D3.1.4] [1]

8 **STATE** the two hormones secreted by the corpus luteum. [D3.1.5] [2]

Section B: Understanding

15 MARKS

9 **COMPARE AND CONTRAST** sexual and asexual reproduction. [D3.1.1] [3]

SEXUAL	ASEXUAL

10 **EXPLAIN** the roles of meiosis and fertilization in a sexual life cycle. [D3.1.2] [3]

11 **EXPLAIN** why male gametes are small and numerous, while female gametes are large and few. [D3.1.3] [3]

12 **OUTLINE** the events of human fertilization, from membrane fusion to the first division. [D3.1.6] [4]

13 **DISTINGUISH** between positive and negative feedback, with one example of each from the menstrual cycle. [D3.1.5] [2]

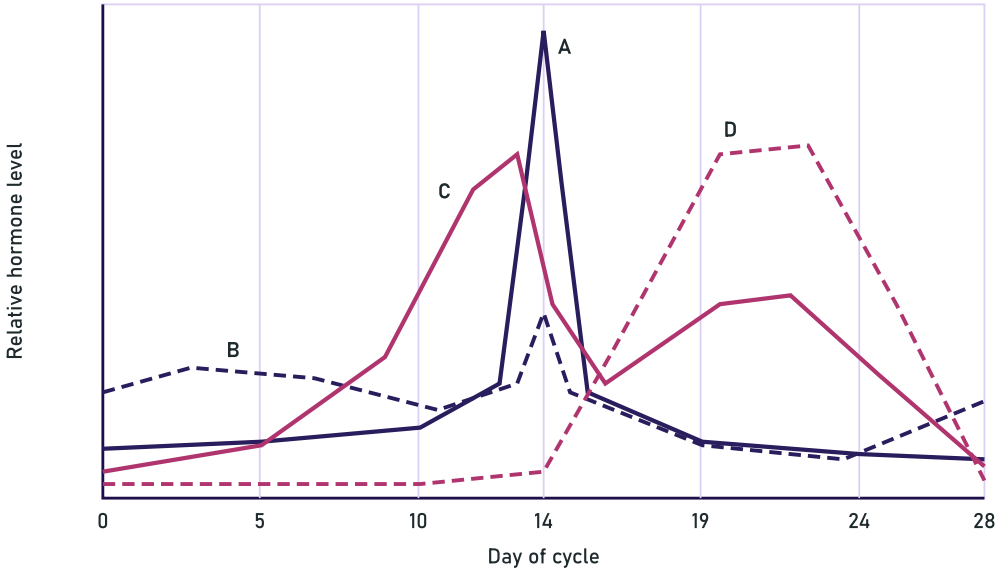
Section C: Working with data

18 MARKS

The graph is a simplified sketch, not measured data. The table in question 15 is invented for practice.

14 [D3.1.5]

The graph shows blood hormone levels during a 28-day menstrual cycle where no pregnancy occurs. Each hormone has its own scale, so only the shapes can be compared.



a IDENTIFY hormones A, B, C and D. [2]

b STATE the day on which ovulation is most likely. [1]

c EXPLAIN what causes the sharp rise in hormone A. [2]

d EXPLAIN why hormone D rises after ovulation. [2]

e EXPLAIN why menstruation begins at the end of the cycle. [2]

f PREDICT how the graph for hormone D would differ after day 24 if an embryo had implanted. [2]

15 [D3.1.7]

The table shows the success of IVF at a fictional clinic.

AGE OF PATIENT / YEARS	UNDER 35	35–37	38–40	41–42	OVER 42
LIVE BIRTHS PER 100 TREATMENTS	32	25	18	10	4

a DESCRIBE the trend. [1]

b CALCULATE the percentage change from the youngest to the oldest group. [2]

c SUGGEST two reasons for the trend, or why a clinic should be cautious about using it to advise one patient. [2]

d Some clinics place one embryo at a time, even though placing more may raise the chance of a pregnancy. SUGGEST why. [2]

Section D: Longer answers

9 MARKS

16 EXPLAIN how hormones are used in IVF treatment. [D3.1.7] [5]

17 DRAW a labelled diagram of the female-typical reproductive system, seen from the front. [D3.1.4] [4]